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COURSE NAME: SOFTWARE ENGINEERING

EXP 5:

Software Requirements Prioritization using MoSCoW Analysis for Library Management System

Aim:

To analyze, prioritize, and categorize the functional requirements of a Library Management System using the MoSCoW method based on user impact and feasibility constraints.

procedure:

- ☐ Open Spreadsheet: Launch Google Sheets or MS Excel and create a new blank sheet.
- ☐ Add Headers: Set Row 1 headers as: Req ID, Requirement Description, MoSCoW Category, Impact on Users, and Feasibility & Justification.
- ☐ List Requirements: Enter all 10 library system requirements into the table.
- ☐ Categorize Requirements: Assign each requirement to Must-Have, Should-Have, Could-Have, or Won't-Have based on operational need and feasibility.
- ☐ Add Justifications: Write a short explanation for each item detailing its user impact and technical effort.
- ☐ Format & Export: Apply bold headers, auto-fit columns, add text wrapping, and save or download as .xlsx or PDF for submission.

Outputs:

B12						
	A	B	C	D	E	F
	Req ID	Requirement Description	MoSCoW Category	Impact on Users / Stakeholders	feasibility	Resource Justification
1	REQ-01	Add a feature to search books by title and author.	Must-Have	high	high	low complexity, low cost; essential for baseline deployment.
2	REQ-02	Implement an online book reservation system.	Must-Have	high	high	standard database transaction logic requiring minimal budget.
3	REQ-03	Generate monthly reports on borrowed books for administrators.	Must-Have	high	high	standard SQL database query and reporting engine function.
4	REQ-04	Enable email notifications for overdue books.	Should-Have	high	high	using standard SMTP mail services with low integration overhead.
5	REQ-05	Create a user-friendly dashboard for librarians.	Should-Have	high	medium	requires UX design and backend API endpoints.
6	REQ-06	Add support for QR code scanning for borrowing and returning books.	Could-Have	medium	medium	requires webcam/mobile integration and hardware testing.
7	REQ-07	Allow users to review and rate books.	Could-Have	medium	high	basic CRUD functionality, but non-essential for initial launch.
8	REQ-08	Provide multi-language support.	Could-Have	medium	medium	requires i18n localization setup and translation management.
9	REQ-09	Integrate a chatbot for user assistance.	Won't-Have	low	low	high complexity, cost, and AI integration efforts for initial release.
10	REQ-10	Develop a mobile app version of the system.	Won't-Have	medium	low	requires cross-platform development, time, and team budget.
11						
12						

Result: Thus, All 10 requirements for the Library Management System were successfully analyzed, categorized, and documented in Google Sheets/Excel using the MoSCoW method. Each requirement was evaluated based on user impact and technical feasibility to establish a clear development priority list.